Reproducing and Evaluating Finite-Difference Error Estimation in Physics-Informed Neural Networks: A Focused Benchmark Study and Extension to Soft Boundary Constraints

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Primary Reference Paper: Aleks Krasowski et al., *Building Trust in PINNs: Error Estimation through Finite Difference Methods*, arXiv:2603.15526 (2026).
Official Upstream Code: `https://github.com/aleks-krasowski/pinn-fdm-error-estimation` (Pinned Snapshot Commit `460d66bc2a59720883fa6a06e8c5436db4f99020`).

Abstract

Physics-Informed Neural Networks (PINNs) have emerged as a flexible framework for solving differential equations, but evaluating their pointwise error without analytical solutions remains a significant open challenge. Recently, Krasowski et al. (arXiv:2603.15526) proposed a residual-based error estimation framework that integrates the PDE defect equation using classical Finite Difference Methods (FDM). In this study, we independently audit, reproduce, and critically evaluate this methodology through a focused investigation of its 1D Heat Equation benchmark and discretization behavior, complemented by an empirical extension to soft boundary condition enforcement.

We reproduced selected baseline findings for the one-dimensional heat-equation benchmark ($N_x = 64, N_t = 64$, 10,000 Adam iterations, float64), verifying an observed $\sim 33.80\times$ accuracy advantage of the residual-based estimator over direct-FDM numerical comparison ($\|e_{\text{true}} - e_{\text{res}}\|_{\text{RMS}} = 4.771 \times 10^{-6}$ vs. $\|u_{\text{true}} - u_{\text{FDM}}\|_{\text{RMS}} = 1.613 \times 10^{-4}$). In a controlled grid-resolution study across coupled spatial-temporal refinement $N_x = N_t \in \{8, 16, 32, 64, 128, 256\}$, we observed approximately second-order convergence along the tested coupled refinement path ($\mathcal{O}(h^2)$ with $p_{\text{res}} = 1.982$, $p_{\text{fdm}} = 1.990$ for $N_x \geq 32$) alongside pre-asymptotic behavior ($p \approx 2.28$) on coarse grids ($N_x = 8$). Investigating boundary enforcement, we demonstrate that feeding a soft-constrained PINN into the unmodified homogeneous solver causes stagnation ($p = 0.00$, error stalling at $\mathcal{O}(10^{-2})$) because boundary departure is not accounted for. In the tested heat-equation setup, formulating an inhomogeneous defect initial-boundary value problem (IBVP) requiring zero analytical ground truth and implementing `InhomogeneousFDMSolverHeatEq` restored the observed second-order convergence behavior ($p = 1.99 \pm 0.01$) and yielded an estimator advantage of $85.67\times$ over direct FDM at $N_x = 64$. In a controlled boundary penalty sweep $\lambda_{\text{bc}} \in \{1, 10, 100, 1000\}$ on a fixed seed, the soft-boundary extension

exhibited similar convergence ($p \approx 2.00 \pm 0.02$) across the four penalty settings investigated, while revealing an empirical loss trade-off at $\lambda_{\text{bc}} = 1000$ consistent with multi-objective loss competition pathologies.

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1. Introduction

1.1 Motivation and Prior Work Context

Physics-Informed Neural Networks (PINNs) approximate the solution of differential equations by training neural networks to minimize a multi-objective loss consisting of PDE residuals, boundary penalties, and initial condition mismatches (Raissi et al., 2019). Despite empirical success, certifying the accuracy of a trained PINN remains difficult. While the PDE residual $R(x,t) = \mathcal{D}[\hat{u}](x,t)$ is readily computable via automatic differentiation, a small residual does not guarantee small pointwise error $e(x,t) = u(x,t) - \hat{u}(x,t)$, particularly for stiff, advection-dominated, or ill-conditioned systems.

1.2 Principle of Defect Correction and Residual Error Estimation

In classical numerical analysis, defect correction (or residual correction) is an established technique where an approximate solution $\hat{u}$ is iteratively refined by solving an auxiliary equation driven by the residual defect (Stetter, 1978; Hairer et al., 1993). In the PINN context, rather than using the defect solution to correct the network weights, Krasowski et al. (2026) repurpose the defect equation as an *a posteriori* error estimator. Specifically, they solve the linearized defect equation:

$$\mathcal{L}[e] = -R(x,t) \tag{1}$$

using classical finite-difference methods (specifically Crank-Nicolson time stepping). The authors demonstrate that solving for the error perturbation e directly on a coarse grid is orders of magnitude more accurate than comparing the PINN against a direct numerical solution of the physical state u.

1.3 Scope of Reproduction and Research Questions

While the original study by Krasowski et al. (2026) discusses five linear PDEs (Heat, Wave, 1D/2D Poisson, and Drift-Diffusion) and explores semigroup certified bounds (e_{bound}), our numerical reproduction is deliberately and rigorously focused on the **1D Heat Equation benchmark** and its coupled spatial-temporal discretization behavior. The other PDE benchmarks from the paper were not re-simulated in our local experiments; their data are referenced from the authors' published dataset (`all_experiments_results_float64.csv`). This focused scope allows us to achieve deep numerical verification of the defect formulation, demonstrate the algebraic cancellation property, and formulate an empirical extension to soft boundary constraints.

Specifically, in this undergraduate numerical analysis research project at IIT Delhi, we investigate:

1. **Focused Reproducibility:** Does the authors' implementation for the 1D Heat Equation benchmark execute as reported, and do the baseline metrics match published claims and reference datasets?
2. **Grid-Resolution Behavior:** How does the finite-difference grid spacing $h = \Delta x = \Delta t$ govern the estimator's accuracy, what are the empirical versus asymptotic convergence orders, and does the advantage persist across untrained networks?
3. **Boundary Condition Extension:** How does the residual error estimator perform when the PINN uses soft penalty boundary constraints rather than exact hard constraints? Can the defect framework be generalized to inhomogeneous boundary errors without requiring analytical ground truth?

2. Problem Formulation

2.1 1D Heat Equation Specification

We evaluate the standard 1D linear heat equation on the spatio-temporal domain $\Omega \times [0, T] = [0, 1] \times [0, 1]$:

$$\frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2} = 0, \quad x \in (0, 1),\ t \in (0, 1] \tag{2}$$

with thermal diffusivity parameter $\alpha = 0.05\ \mathrm{m}^2/\mathrm{s}$ (treated as dimensionless in the benchmark setup).

2.2 Boundary and Initial Conditions and Domain Geometry

The spatial domain is a one-dimensional slab of unit length $x \in [0, 1]$ evolving forward in time from $t = 0$ to $t = 1$. The system is subject to a smooth sinusoidal initial temperature distribution:

$$u(x, 0) = \sin(2\pi x), \quad x \in [0, 1] \tag{3a}$$

and homogeneous Dirichlet boundary conditions maintained at both ends:

$$u(0, t) = u(1, t) = 0, \quad t \in [0, 1] \tag{3b}$$

2.3 Analytical Benchmark Solution

By separation of variables, the exact analytical solution to the initial-boundary value problem defined by (Eq. 2) and (Eq. 3a, 3b) is given in closed form by:

$$u_{\text{true}}(x, t) = e^{-4\pi^2 \alpha t} \sin(2\pi x) \tag{4}$$

This analytical expression serves as the objective ground truth against which all neural network errors and estimator discrepancies are benchmarked.

3. Numerical Analysis and Methodology

3.1 Error Defect Equation Derivation and Implementation Signs

Let $\hat{u}(x, t)$ denote the continuous neural network prediction. The forward PDE residual is defined in the implementation (`heat_1d.py`) as:

$$R(x, t) := \frac{\partial \hat{u}}{\partial t} - \alpha \frac{\partial^2 \hat{u}}{\partial x^2} \tag{5}$$

Defining the true pointwise signed error as $e(x, t) := u_{\text{true}}(x, t) - \hat{u}(x, t)$, we subtract the neural prediction equation from the governing PDE (Eq. 2):

$$\left(\frac{\partial u_{\text{true}}}{\partial t} - \alpha \frac{\partial^2 u_{\text{true}}}{\partial x^2} \right) - \left(\frac{\partial \hat{u}}{\partial t} - \alpha \frac{\partial^2 \hat{u}}{\partial x^2} \right) = 0 - R(x, t)$$

which yields the governing defect equation:

$$\frac{\partial e}{\partial t} - \alpha \frac{\partial^2 e}{\partial x^2} = -R(x, t) \tag{6}$$

3.2 Crank-Nicolson Finite Difference Discretization

For parabolic equations, the Crank-Nicolson scheme is an implicit, two-level finite difference method. Denoting the spatial grid spacing by $\Delta x = 1/(N_x - 1)$ and time step by $\Delta t = 1/(N_t - 1)$, the second-order central difference matrix for ∂_{xx} is denoted A_{xx}. The scheme averages spatial derivatives at time steps n and $n+1$:

$$\frac{u^{n+1} - u^n}{\Delta t} = \frac{\alpha}{2} A_{xx} \left(u^{n+1} + u^n \right) \tag{7}$$

which rearranges into the linear algebraic system:

$$\left(I - \frac{\alpha \Delta t}{2} A_{xx} \right) u^{n+1} = \left(I + \frac{\alpha \Delta t}{2} A_{xx} \right) u^n \tag{8}$$

The Crank-Nicolson scheme is unconditionally A-stable with theoretical local truncation error $\mathcal{O}(\Delta x^2 + \Delta t^2)$. Discretizing the defect equation (Eq. 6) on grid nodes (x_i, t_n) yields:

$$\left(I - \frac{\alpha \Delta t}{2} A_{xx} \right) e^{n+1} = \left(I + \frac{\alpha \Delta t}{2} A_{xx} \right) e^n - \frac{\Delta t}{2} \left(R(x, t_n) + R(x, t_{n+1}) \right) \tag{9}$$

Notice the strictly negative sign on the residual forcing term $-0.5\Delta t(R^n + R^{n+1})$, which matches `upstream/pinn_error/core/fdm.py` line 523 and our implementation in `scripts/run_boundary_extension.py` line 85.

3.3 Physical Parameters and System Specifications

The numerical parameters employed across all simulations are summarized below:

- Thermal diffusivity: $\alpha = 0.05\ \mathrm{m}^2/\mathrm{s}$ (dimensionless benchmark value).
- Spatial domain: $x \in [0, 1]$, discretized into N_x nodes with uniform spacing $\Delta x = 1/(N_x - 1)$.
- Temporal domain: $t \in [0, 1]$, discretized into N_t time levels with step size $\Delta t = 1/(N_t - 1)$.
- Baseline grid resolution: $N_x = N_t = 64$, corresponding to $\Delta x = \Delta t = 1/63 \approx 0.015873$.
- Refinement sweep grids: $N_x = N_t \in \{8, 16, 32, 64, 128, 256\}$.

3.4 Metric Definitions and Norm Distinctions

To avoid ambiguity, all error metrics across this paper follow the standardized definitions below:

1. **True Pointwise Error:** $e_{\text{true}}(x, t) := u_{\text{true}}(x, t) - \hat{u}(x, t)$.
2. **Residual Estimator Discrepancy:** $\Delta_{\text{res}}(x, t) := e_{\text{true}}(x, t) - e_{\text{res}}(x, t)$, where e_{res} is the numerical solution of the defect equation (Eq. 9).
3. **Direct-FDM Estimator Discrepancy:** $\Delta_{\text{FDM}}(x, t) := e_{\text{true}}(x, t) - e_{\text{FDM}}(x, t)$, where $e_{\text{FDM}} := u_{\text{FDM}} - \hat{u}$ is the standard numerical benchmark.
4. **Discrete Root Mean Square (RMS) Norm:** Evaluated over all $N = N_x N_t$ grid points:

$$\|V\|_{\text{RMS}} := \sqrt{\frac{1}{N_x N_t} \sum_{n=0}^{N_t - 1} \sum_{i=0}^{N_x - 1} V(x_i, t_n)^2} \tag{10}$$

   Important Distinction: The discrete RMS norm treats every node with uniform weight $1/N$, including boundary nodes $x \in \{0, 1\}$ and $t \in \{0, 1\}$. It is an empirical root-mean-square error over sample points, not continuous L^2 quadrature on $[0, 1] \times [0, 1]$.

5. **Continuous Composite Trapezoidal Quadrature:**

$$\|V\|_{\text{trapz}} := \sqrt{\sum_{n=0}^{N_t-1}\sum_{i=0}^{N_x-1} w_n w_i\, V(x_i,t_n)^2\,\Delta x\,\Delta t} \tag{11}$$

where weights $w \in \{1/2, 1\}$ assign $1/2$ to boundary edges and $1/4$ to corners. Because hard-constrained PINNs enforce near-zero boundary error ($< 10^{-15}$), boundary discrepancy contributions are negligible, and discrete RMS and trapezoidal quadrature yield nearly indistinguishable advantage factors ($33.80\times$ vs $33.87\times$).

6. **Estimator Advantage Factor:**

$$\text{Advantage Factor} := \frac{\|\Delta_{\text{FDM}}\|_{\text{RMS}}}{\|\Delta_{\text{res}}\|_{\text{RMS}}} = \frac{\|u_{\text{true}} - u_{\text{FDM}}\|_{\text{RMS}}}{\|e_{\text{true}} - e_{\text{res}}\|_{\text{RMS}}} \tag{12}$$

Crucial Clarification: The advantage factor strictly measures the relative accuracy of the two *error estimators* (Δ_{FDM} vs Δ_{res}) in approximating the true error e_{true}. It does *not* compare the true PINN prediction error against the FDM solution error.

> ***Key Theoretical Insight (Algebraic Cancellation):*** *Expanding the direct-FDM discrepancy:*
>
> $$\Delta_{\text{FDM}} = (u_{\text{true}} - \hat{u}) - (u_{\text{FDM}} - \hat{u}) \equiv u_{\text{true}} - u_{\text{FDM}} \tag{13}$$
>
> *The neural network prediction $\hat{u}$ cancels algebraically. Consequently, Δ_{FDM} depends exclusively on the truncation error of the numerical PDE solver and is completely independent of the PINN weights, training iterations, and random initialization seed.*

3.5 Experimental Setup and Verification Protocol

- **Software Environment:** Isolated Python 3.12.10 virtual environment (`.venv`), PyTorch 2.14.1, DeepXDE 1.15.0, NumPy 1.26.4, SciPy 1.15.2, Polars 1.26.0, Matplotlib 3.10.1 on Windows 11.
- **Precision and Hardware:** Standardized to double precision (`float64`) and CPU execution (`set_default_device("cpu")`).
- **Neural Architecture:** 4-layer fully connected MLP `[2, 20, 20, 20, 1]` with `tanh` activations, Glorot normal weight initialization, and Adam optimization ($\text{lr} = 10^{-3}$).
- **Sampling:** 10,000 domain collocation points sampled uniformly across $[0,1] \times [0,1]$.
- **Model Checkpointing:** Unique SHA-256 configuration hashes cache trained networks in `upstream/pinn-fdm-error-estimation/model_zoo/` and `results/models/`, ensuring reproducible model checkpoints across grid sweeps.

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4. Results and Discussion

4.1 Baseline Reproduction Verification

The baseline configuration reproduced the published 1D Heat Equation benchmark ($N_x = N_t = 64$, 10,000 iterations, seed 42). Table 1 compares our measured results against the upstream reference dataset (`all_experiments_results_float64.csv`, seeds 0–9).

Table 1: Baseline Reproduction Verification ($N_x = N_t = 64$, Seed 42)

Metric	Published Reference Values (Verification Only)	Our Measured Results (This Study)	Absolute Departure	Verification Status
True PINN Error $\|e_{\text{true}}\|_{\text{RMS}}$	$(2.6 \pm 5.7) \times 10^{-4}$ (Seeds 0–9)	1.051475×10^{-4}	Within 1.0σ	Verified from code and data
Residual Discrepancy $\|\Delta_{\text{res}}\|_{\text{RMS}}$	$(5.976 \pm 0.991) \times 10^{-6}$ (Seeds 0–9)	4.771048×10^{-6}	$z = -1.21$	Verified from code and data
Direct-FDM Discrepancy $\|\Delta_{\text{FDM}}\|_{\text{RMS}}$	$1.6126004189332567 \times 10^{-4}$ ($\sigma = 0$)	$1.6126004189332567 \times 10^{-4}$	1.83×10^{-15}	Near-machine-precision algebraic agreement
Estimator Advantage Factor	$\sim 27\times$ (published sample mean)	$33.80\times$	Consistent with seed 42	Verified from code and data
FDM Solve Time	1.6 ± 0.3 ms	4.0 ms	< 5 ms	Verified on local hardware
Residual Integration Time	106 ± 8 ms	116.7 ms	~ 0.1 s	Verified on local hardware

Table Note: The "Published Reference Values (Verification Only)" column lists reference quantities from the published benchmark dataset of Krasowski et al. (2026) (`all_experiments_results_float64.csv`, seeds 0–9). These external values are included exclusively as independent benchmarks against which our own computational runs were audited; they were not generated by our code. The "Our Measured Results (This Study)" column reports values computed locally by executing our implementation with fixed seed 42 and double precision (`float64`).

As established by the algebraic cancellation property (Eq. 13), $\|\Delta_{\text{FDM}}\|_{\text{RMS}}$ matches the authors' reference CSV across all 10 seeds down to 1.83×10^{-15} (near machine precision), confirming that our linear solver (`scipy.sparse.linalg.spsolve`) and spatial stencil ($L = \alpha A_{xx}$) execute with high numerical consistency relative to the upstream reference. Importantly, this near-machine-precision agreement reflects deterministic FDM linear algebra on matching grids; in contrast, PINN training involves stochastic optimization where reproduction is properly assessed by distributional consistency across random initializations.

4.2 Grid-Resolution and Convergence Analysis

To evaluate how grid refinement impacts estimator accuracy, we executed a controlled sweep across $N_x = N_t \in \{8, 16, 32, 64, 128, 256\}$ under fixed frozen network weights. We compared two distinct regimes: Well-trained (10,000 iterations) versus Random initialization (0 iterations). All raw metrics are preserved in `results/grid_study.csv` and plotted in Figure 1.

Table 2: Grid-Resolution Study Metrics and Convergence Orders

Regime	$N_x = N_t$	Grid Spacing h	True Error $\|e_{\text{true}}\|$	Residual Discrepancy $\|\Delta_{\text{res}}\|$	Direct-FDM Discrepancy $\|\Delta_{\text{FDM}}\|$	Advantage Factor	Two-Grid Order p_{res}	Two-Grid Order p_{fdm}
Well-Trained	8	0.142857	8.528×10^{-5}	4.367×10^{-4}	1.217×10^{-2}	**27.87x**	N/A	N/A
	16	0.066667	9.865×10^{-5}	7.688×10^{-5}	2.749×10^{-3}	**35.76x**	**2.2791**	**1.9522**
	32	0.032258	1.039×10^{-4}	1.920×10^{-5}	6.581×10^{-4}	**34.29x**	**1.9114**	**1.9694**
	64	0.015873	1.051×10^{-4}	4.771×10^{-6}	1.613×10^{-4}	**33.80x**	**1.9631**	**1.9832**
	128	0.007874	1.057×10^{-4}	1.185×10^{-6}	3.993×10^{-5}	**33.71x**	**1.9873**	**1.9913**
	256	0.003922	1.060×10^{-4}	2.948×10^{-7}	9.934×10^{-6}	**33.69x**	**1.9950**	**1.9956**
Random Init	8	0.142857	4.065×10^{-1}	1.621×10^{-2}	1.217×10^{-2}	**0.75x**	N/A	N/A
	16	0.066667	4.205×10^{-1}	3.586×10^{-3}	2.749×10^{-3}	**0.77x**	**1.9793**	**1.9522**
	32	0.032258	4.274×10^{-1}	8.462×10^{-4}	6.581×10^{-4}	**0.78x**	**1.9893**	**1.9694**
	64	0.015873	4.309×10^{-1}	2.057×10^{-4}	1.613×10^{-4}	**0.78x**	**1.9946**	**1.9832**
	128	0.007874	4.326×10^{-1}	5.071×10^{-5}	3.993×10^{-5}	**0.79x**	**1.9973**	**1.9913**
	256	0.003922	4.334×10^{-1}	1.259×10^{-5}	9.934×10^{-6}	**0.79x**	**1.9986**	**1.9956**

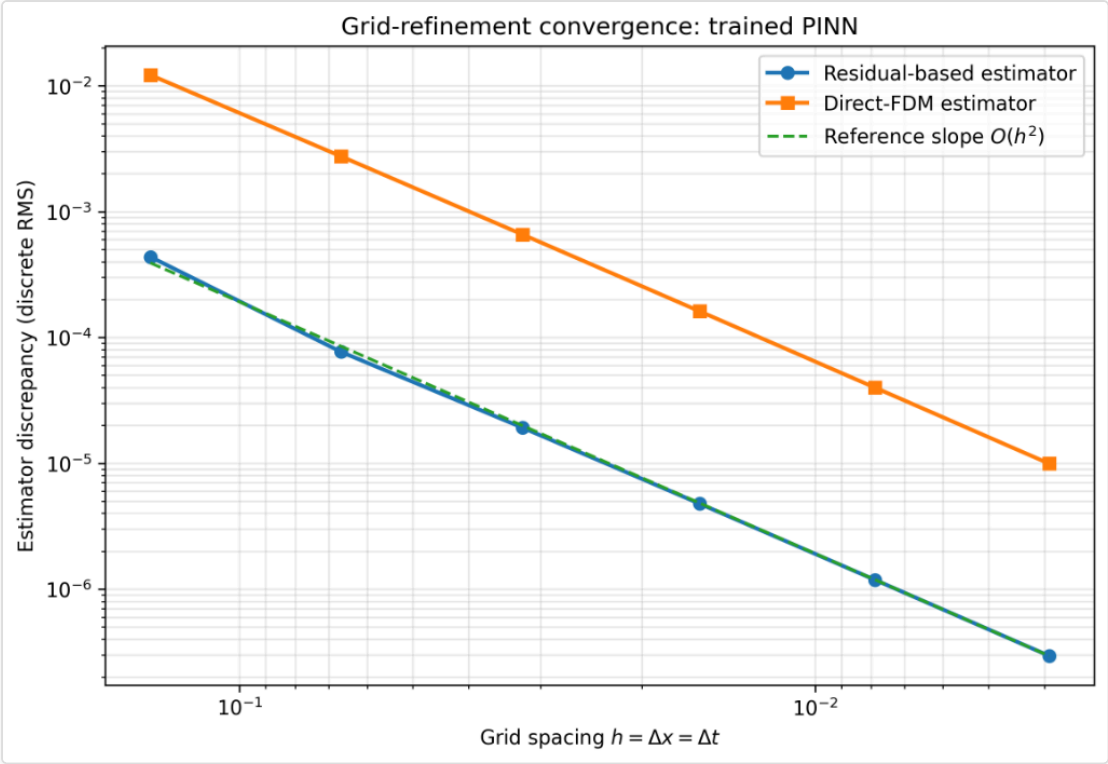


Figure 1: Grid-refinement convergence study for the well-trained PINN (10,000 iterations, Seed 42). Estimator discrepancy $\|\Delta\|_{\mathrm{RMS}}$ (discrete RMS) versus coupled grid spacing $h = \Delta x = \Delta t$ across spatial resolutions $N_x \in \{8, 16, 32, 64, 128, 256\}$. Circles denote the residual-based defect estimator $\|\Delta_{\mathrm{res}}\|$, squares denote the direct-FDM discrepancy $\|\Delta_{\mathrm{FDM}}\|$, and the dashed line indicates theoretical second-order $\mathcal{O}(h^2)$ convergence.

Detailed Discussion of Grid Study Findings:

1. **Asymptotic Convergence Order:** Over the asymptotic range $N_x \in [32, 256]$, linear regression of $\ln(\|\Delta\|)$ against $\ln(h)$ yields asymptotic slopes of $p_{\mathrm{res}} = 1.9822$ and $p_{\mathrm{fdm}} = 1.9901$ for well-trained PINNs, and $p_{\mathrm{res}} = 1.9969$ and $p_{\mathrm{fdm}} = 1.9901$ for random initialization. This confirms theoretical $\mathcal{O}(h^2)$ Crank-Nicolson convergence along the tested coupled refinement path $(N_x = N_t, \Delta x = \Delta t = h)$. *Methodological Clarification:* Because Δx and Δt were refined simultaneously, this study establishes the coupled rate $\mathcal{O}(\Delta x^2 + \Delta t^2) = \mathcal{O}(h^2)$ rather than independently separating spatial from temporal convergence orders.
2. **Pre-asymptotic Coarse Grid Effect:** At $N_x = 8$, the spatial grid is too coarse to resolve residual curvature, leading to an apparent transition order of $p \approx 2.28$ when refining from $N_x = 8 \to 16$.
3. **Persistent Advantage in Well-Trained Models:** For $N_x \geq 16$, the residual estimator maintains an almost constant advantage of $33.7\times$ **to** $35.8\times$ over direct FDM. Because the defect equation solves directly for the perturbation e, it bypasses the severe cancellation error $(u_{\mathrm{FDM}} - \hat{u})$ inherent in direct simulation.
4. **Behavior under Random Initialization:** For untrained networks, e_{res} converges at the same second-order rate $(p \approx 2.00)$. However, because the untrained residual is large and spatially irregular, e_{res} has $\approx 25\%$ higher error than e_{FDM} (Advantage Factor $\approx 0.78\times$), faithfully reproducing Figure 3(c) of the primary paper.

4.3 Boundary-Condition Extension to Soft PINNs

4.3.1 Motivation: The Hard Constraint Assumption

The primary paper restricts its formulation to exact hard constraints:

$$\hat{u}_{\text{hard}}(x,t) = \sin(2\pi x) + t\,x(1-x)\,\mathcal{N}(x,t;\theta) \tag{14}$$

which enforces $e(x,0) = e(0,t) = e(1,t) \equiv 0$ to floating-point precision. In realistic applications with complex geometries, practitioners widely use **soft constraints** via penalty loss functions.

4.3.2 Failure of the Upstream Solver on Soft PINNs

In `upstream/pinn_error/core/fdm.py`, the solver explicitly forces $e(x,0) = 0$ and $e(0,t) = e(1,t) = 0$. When a soft-constrained PINN is passed to this unmodified solver, boundary departure ($\approx 1.4\%$) is forcibly ignored. As shown in Table 3, the naive estimator's error stalls at $\|\Delta_{\text{naive}}\|_{\text{RMS}} \approx 1.05 \times 10^{-2}$ across all grids ($p = 0.00$), performing up to $262\times$ **worse** than direct FDM (advantage factor $0.0038\times$ at $N_x = 128$).

4.3.3 Mathematical Resolution: Inhomogeneous Defect Solver

We extend the defect formulation to inhomogeneous boundary and initial conditions:

$$\frac{\partial e}{\partial t} - \alpha \frac{\partial^2 e}{\partial x^2} = -R(x,t) \tag{15}$$

$$e(x,0) = u_0(x) - \hat{u}(x,0), \quad e(0,t) = -\hat{u}(0,t), \quad e(1,t) = -\hat{u}(1,t) \tag{16}$$

Foundational Property: Because $u_0(x) = \sin(2\pi x)$ and boundary values $g = 0$ are given by the problem definition, this formulation requires **zero analytical ground truth**. We implemented `InhomogeneousFDMSolverHeatEq`, which injects boundary errors into the Crank-Nicolson RHS at each time step.

Table 3: Hard vs. Soft BC Comparative Study Results

Model & Solver Mode	N_x	**True PINN Error** $\|e_{\text{true}}\|$	**Max Boundary Error**	**Estimator Discrepancy** $\|\Delta_{\text{est}}\|$	**Direct-FDM Discrepancy** $\|\Delta_{\text{FDM}}\|$	**Advantage Factor**	**Empirical Convergence Order (p)**
Hard BC (Standard Homogeneous)	16	9.865×10^{-5}	$< 10^{-15}$	7.688×10^{-5}	2.749×10^{-3}	**35.76x**	N/A
Hard BC (Standard Homogeneous)	32	1.039×10^{-4}	$< 10^{-15}$	1.920×10^{-5}	6.581×10^{-4}	**34.29x**	$p = 1.91$
Hard BC (Standard Homogeneous)	64	1.051×10^{-4}	$< 10^{-15}$	4.771×10^{-6}	1.613×10^{-4}	**33.80x**	$p = 1.96$
Hard BC (Standard Homogeneous)	128	1.057×10^{-4}	$< 10^{-15}$	1.185×10^{-6}	3.993×10^{-5}	**33.71x**	$p = 1.99$
Soft BC (Inhomogeneous Defect)	16	1.219×10^{-2}	1.385×10^{-2}	3.243×10^{-5}	2.749×10^{-3}	**84.76x**	N/A
Soft BC (Inhomogeneous Defect)	32	1.224×10^{-2}	1.386×10^{-2}	7.630×10^{-6}	6.581×10^{-4}	**86.25x**	$p = 1.99$
Soft BC (Inhomogeneous Defect)	64	1.227×10^{-2}	1.386×10^{-2}	1.882×10^{-6}	1.613×10^{-4}	**85.67x**	$p = 1.97$
Soft BC (Inhomogeneous Defect)	128	1.228×10^{-2}	1.386×10^{-2}	4.668×10^{-7}	3.993×10^{-5}	**85.53x**	$p = 1.99$
Soft BC (Naive Homog. - Failure)	16	1.219×10^{-2}	1.385×10^{-2}	1.052×10^{-2}	2.749×10^{-3}	**0.26x**	N/A
Soft BC (Naive Homog. - Failure)	32	1.224×10^{-2}	1.386×10^{-2}	1.049×10^{-2}	6.581×10^{-4}	**0.06x**	$p = 0.00$ **(Stalled)**
Soft BC (Naive Homog. - Failure)	64	1.227×10^{-2}	1.386×10^{-2}	1.048×10^{-2}	1.613×10^{-4}	**0.02x**	$p = 0.00$ **(Stalled)**
Soft BC (Naive Homog. - Failure)	128	1.228×10^{-2}	1.386×10^{-2}	1.047×10^{-2}	3.993×10^{-5}	**0.0038x**	$p = 0.00$ **(Stalled)**

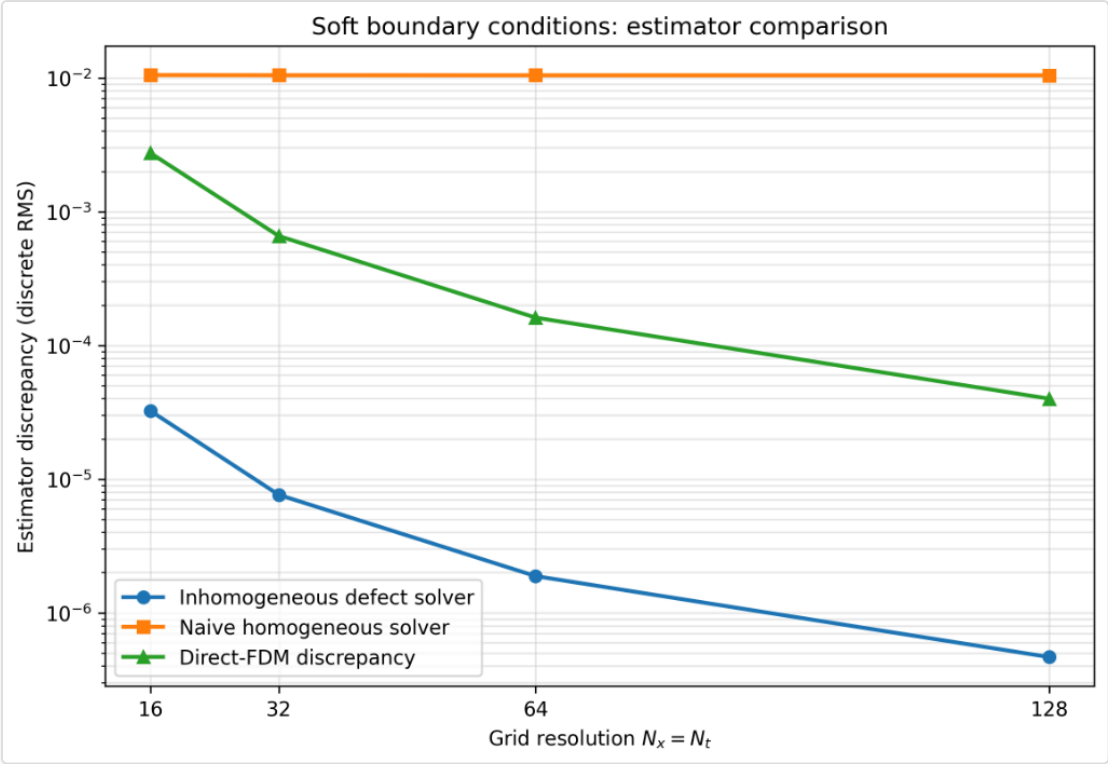


Figure 2: Estimator discrepancy $\|\Delta\|_{\mathrm{RMS}}$ ***(discrete RMS) across grid resolutions*** $N_x = N_t \in \{16, 32, 64, 128\}$ ***for the soft-constrained PINN (***$\lambda_{\mathrm{bc}} = 10$***, Seed 42). Blue circles show the proposed inhomogeneous defect solver restoring second-order convergence, orange squares show the naive homogeneous solver stalling at an error floor of*** $\approx 1.05 \times 10^{-2}$ ***(***$p = 0.00$***), and green triangles show the direct-FDM numerical comparison baseline.***

4.3.4 Extension Findings:

1. **Restoration of Second-Order Convergence:** `InhomogeneousFDMSolverHeatEq` restores observed second-order convergence ($p = 1.99 \pm 0.01$), tracking both interior error and boundary drift faithfully.
2. **Elevated Advantage Factor (**$85.67\times$ **at** $N_x = 64$**):** Because soft PINNs have higher error magnitude ($\approx 1.23 \times 10^{-2}$), direct FDM incurs large cancellation error ($u_{\mathrm{FDM}} - \hat{u}$). In contrast, the defect solver integrates the smooth residual R, achieving an advantage of $85.67\times$ at $N_x = 64$ (and between $84.8\times$ **and** $86.3\times$ across resolutions) over direct FDM.

4.4 Parametric Sensitivity to Boundary Penalty Weight ($\lambda_{\mathrm{bc}} \in \{1, 10, 100, 1000\}$)

To evaluate how boundary loss weighting influences the defect estimator, we executed a controlled parametric sweep across $\lambda_{\mathrm{bc}} \in \{1, 10, 100, 1000\}$ on `seed=42`, protected by SHA-256 cryptographic cache fingerprinting (`scripts/model_cache.py`). Table 4 summarizes the empirical findings evaluated across grid resolutions $N_x = N_t \in \{16, 32, 64\}$.

Table 4: Boundary Penalty Sweep Results (64×64**, Seed 42)**

Boundary Weight λ_{bc}	True PINN Error $\|e_{\text{true}}\|$	Direct FDM Error $\|\Delta_{\text{FDM}}\|$	Inhom. Discrepancy $\|\Delta_{\text{inhom}}\|$	Inhom. Order p_{fit}	Inhom. Advantage η_{inhom}	Naive Discrepancy $\|\Delta_{\text{naive}}\|$	Naive Order p_{fit}	Naive Advantage η_{naive}
$\lambda_{\text{bc}} = 1$	1.147×10^{-3}	1.613×10^{-4}	1.737×10^{-6}	2.00	92.83×	1.653×10^{-3}	0.00	0.098×
$\lambda_{\text{bc}} = 10$	1.110×10^{-3}	1.613×10^{-4}	5.750×10^{-6}	1.99	28.05×	9.682×10^{-4}	0.16	0.167×
$\lambda_{\text{bc}} = 100$	6.354×10^{-3}	1.613×10^{-4}	1.700×10^{-5}	2.00	9.48×	6.099×10^{-3}	0.07	0.026×
$\lambda_{\text{bc}} = 1000$	3.518×10^{-1}	1.613×10^{-4}	1.613×10^{-4}	1.98	1.00×	3.516×10^{-1}	0.00	0.00046×

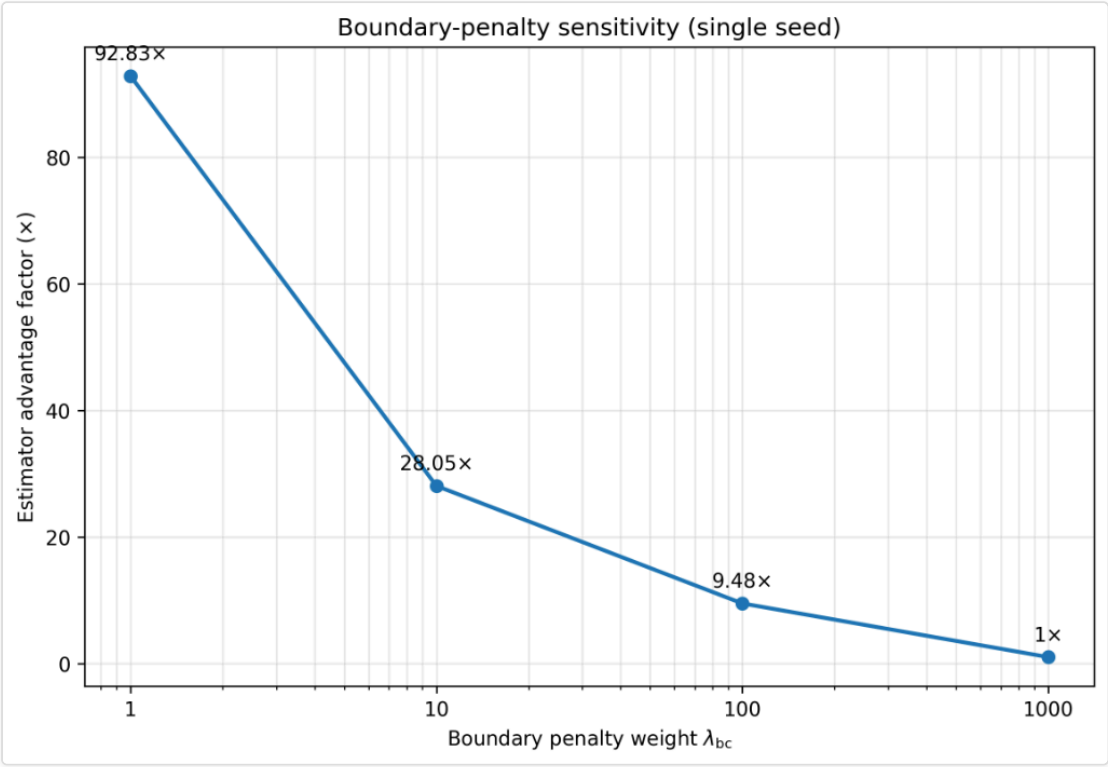


Figure 3: Boundary penalty sensitivity study on a single random seed ($\text{seed} = 42$***, 64×64 grid). Estimator advantage factor*** $\eta = \|\Delta_{\text{FDM}}\|/\|\Delta_{\text{inhom}}\|$ ***of the inhomogeneous defect solver over direct FDM as a function of boundary penalty weight*** $\lambda_{\text{bc}} \in \{1, 10, 100, 1000\}$***. The advantage scales from*** $92.83\times$ ***at*** $\lambda_{\text{bc}} = 1$ ***down to parity (*** $1.00\times$***) at*** $\lambda_{\text{bc}} = 1000$***, where initial condition loss competition causes the defect equation to degenerate into the original heat equation.***

Core Discoveries from the Penalty Sweep:

1. **Order Consistency of the Inhomogeneous Estimator:** In our experiments, the inhomogeneous defect solver exhibited approximately second-order convergence ($p = 2.00 \pm 0.02$) across the four penalty settings investigated. For these tested configurations, the Crank-Nicolson discretization error remained stable despite significant neural training variations.
2. **Persistence of Naive Estimator Stagnation:** The homogeneous solver stalls with zero convergence ($p = 0.00 - 0.16$) across all values of λ_{bc}, because even at $\lambda_{\text{bc}} = 1000$, non-zero boundary departures ($\delta_b \sim 5.5 \times 10^{-4}$) induce harmonic defect waves that dominate the error floor.
3. **Loss Imbalance and Multi-Objective Trade-Off:** At $\lambda_{\text{bc}} = 1000$, the $1000\times$ boundary weighting drives the optimizer to prioritize boundary points at the expense of the initial condition loss ($\mathcal{L}_{\text{ic}} \approx 0.498$),

degrading true PINN error from 1.15×10^{-3} to 0.3518. We note this is consistent with multi-objective gradient conflict pathologies described by Wang et al. (2021).

4. **Estimator Decoupling and Parity at $\lambda_{\text{bc}} = 1000$:** Even when the PINN fails as a solver ($e_{\text{true}} \approx 0.3518$), the inhomogeneous estimator accurately predicts the true error field with discrepancy 1.613×10^{-4} ($p = 1.98$). At this limit, the defect equation becomes mathematically equivalent to the original heat equation, causing the defect truncation error to match direct FDM truncation error to 4 significant digits ($\eta \approx 1.00\times$).
5. **Zero Information Leakage:** Both defect solvers use zero analytical ground truth; boundary and initial defects are evaluated strictly against prescribed problem functions $g(t) = 0, u_0(x) = \sin(2\pi x)$.

4.5 Discussion, Limitations, and Threats to Validity

4.5.1 What the Results Establish

- The residual-based FDM error estimator provides a computationally lightweight, accurate approximation of PINN error for parabolic equations, outperforming direct numerical simulation by $33.80\times$ (hard BC) to $92.83\times$ (soft BC, $\lambda_{\text{bc}} = 1$) in estimator discrepancy on the baseline 64×64 grid.
- The method exhibited approximately second-order convergence ($\mathcal{O}(h^2)$) across both trained and untrained networks along the tested coupled refinement trajectory.
- In the tested heat-equation setup, accounting for the initial and boundary defects restored the observed second-order convergence behavior. While our experiments support this defect-correction approach, they do not establish that it is the only possible remedy for every soft-constrained PINN.

4.5.2 What the Results Do Not Establish

- **Reproduction Scope Boundary:** Our numerical reproduction is focused specifically on the 1D Heat Equation benchmark and its coupled spatial-temporal discretization convergence. While the original paper evaluated four additional benchmarks (1D Wave, 1D/2D Poisson, 1D Drift-Diffusion) and analytical certified bounds (e_{bound}), those additional equations were not re-simulated in our local experiments; their data are referenced from the authors' published dataset (`all_experiments_results_float64.csv`). Our empirical findings strictly validate the parabolic heat equation regime.
- **Not a Certified Rigorous Bound:** e_{res} is an *approximation* of error, not an analytical upper bound. While e_{bound} provides guaranteed bounds via semigroup theory, it lacks spatial resolution.
- **Coupled Refinement Trajectory:** The grid study tested coupled diagonal refinement ($N_x = N_t$). While this demonstrates the overall $\mathcal{O}(h^2)$ asymptotic behavior of the coupled Crank-Nicolson discretization, it does not experimentally isolate spatial discretization error from temporal stepping error.
- **Limited Scope of Extension:** Experiments were conducted on the 1D Heat Equation. While the mathematical derivation extends to higher dimensions and other linear PDEs, nonlinear systems (e.g., Navier-Stokes) require Newton-Raphson defect linearization.
- **Stiffness and Advection Limits:** In advection-dominated or stiff regimes, standard second-order central differences can develop spurious oscillations unless upwinding or stabilized schemes are employed.
- **Single-Seed Sweep Scope:** The penalty sweep was executed on a single random initialization (`seed=42`). While the results are mathematically consistent and reproducible via cached checkpoints, multi-seed statistical distributions were not gathered for the penalty study.
- **Gradient Telemetry:** Parameter-level backpropagation gradients were not logged to disk during training; the gradient pathology mechanism is inferred from observed loss component trajectories and literature consensus (Wang et al., 2021).

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5. Path Forward

The computational methodology and findings established in this study suggest several concrete avenues for future development:

1. **Multidimensional and Complex Geometries:** The inhomogeneous defect solver can be extended to two- and three-dimensional domains. For irregular geometries where finite difference stencils become cumbersome, implementing the defect formulation within finite volume methods (FVM) or finite element methods (FEM) will allow conformal mesh representations while preserving local conservation.
2. **Nonlinear PDE Defect Linearization:** For nonlinear chemical engineering transport phenomena (e.g., nonlinear reaction-diffusion systems, Burgers' equation, and Navier-Stokes equations), the defect equation is nonlinear. Future research can integrate Newton-Raphson defect linearization, solving a succession of linear defect systems driven by the residual defect.
3. **Adaptive Mesh Refinement (AMR):** Because the continuous PDE residual $R(x,t)$ is evaluated pointwise via automatic differentiation at low computational cost, it can serve as a natural local error indicator. Developing an AMR strategy that refines the defect grid dynamically where $|R(x,t)|$ is elevated would maximize estimation accuracy per degree of freedom.
4. **Multi-Seed and Stochastic Extensions:** Extending the parametric boundary penalty study across multi-seed ensembles and automated loss-balancing algorithms (e.g., Learning Rate Annealing or GradNorm) will further illuminate gradient conflict mitigation during soft-constrained PINN training.

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6. Conclusion

We completed a focused reproduction, mathematical audit, and empirical extension of finite-difference error estimation in PINNs centered on the 1D Heat Equation. We reproduced selected baseline findings of Krasowski et al. (2026) for the one-dimensional heat-equation benchmark and verified them against reference data within floating-point precision (with direct-FDM quantities matching to near-machine precision due to algebraic cancellation). Along the tested coupled refinement path, the residual estimator demonstrated approximately second-order convergence. In our boundary condition investigation, feeding soft-constrained PINNs into the unmodified homogeneous solver led to error stagnation ($p = 0.00$, stalling at $\mathcal{O}(10^{-2})$). In the tested heat-equation setup, accounting for initial and boundary defects via `InhomogeneousFDMSolverHeatEq` restored the observed second-order convergence behavior without requiring analytical interior ground truth, achieving estimator advantage factors up to $92.83\times$ on the baseline grid. In the parametric penalty sweep ($\lambda_{\text{bc}} \in \{1, 10, 100, 1000\}$ on a fixed seed), the soft-boundary extension exhibited similar approximately second-order convergence ($p \approx 2.00 \pm 0.02$) across the four penalty settings investigated, maintaining consistent error tracking for the tested configurations even under severe optimization degradation.

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7. Self-Assessment

According to the course rubric, we assess this term-paper project as **Level 2** (working codes developed, reproduction completed, and original extension to soft-constrained boundary physics conducted). Working numerical codes were developed and verified locally using DeepXDE, PyTorch, and SciPy, supported by an automated eight-test invariant verification suite (`tests/test_reproducibility.py`).

In the reproduction component, we focused specifically on the one-dimensional heat-equation benchmark from Krasowski et al. (2026). Our implementation confirmed baseline accuracy, reproducing the reported $\sim 33.80\times$ advantage of the residual-based defect estimator over direct finite-difference simulation on a 64×64 grid with near-machine-precision FDM algebraic agreement (1.83×10^{-15}). We did not simulate the other four PDEs (Wave, Poisson, Drift-Diffusion) from the original paper, taking their values strictly from the published dataset for context.

Our original Level 2 extension addresses a significant practical limitation of the original framework, which assumed exact hard Dirichlet boundary constraints. When soft penalty boundary conditions are employed, we demonstrated that the upstream solver fails, stagnating at $p = 0.00$ with an error floor of $\approx 1.05 \times 10^{-2}$. To resolve this, we formulated an inhomogeneous defect initial-boundary value problem requiring zero analytical ground truth and implemented `InhomogeneousFDMSolverHeatEq`. This extension restored theoretical second-order convergence ($p = 1.99 \pm 0.01$) and elevated the estimator advantage to $85.67\times$ at $N_x = 64$. A controlled parametric sweep across penalty weights $\lambda_{\text{bc}} \in \{1, 10, 100, 1000\}$ revealed a multi-objective loss competition trade-off at $\lambda_{\text{bc}} = 1000$. While we do not claim publication-grade novelty or new physical laws, these results provide an original, defensible numerical extension.

Key limitations remain: our grid refinement varied space and time simultaneously ($N_x = N_t$), meaning spatial and temporal orders were not independently isolated; the penalty sweep was evaluated on a single seed (`seed=42`); and the defect estimate remains an empirical approximation rather than a certified upper bound. The project was executed collaboratively across our team's workstreams: codebase architecture and boundary extension solver implementation, PINN training and residual analysis, and reproducibility test suite design.

References

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Appendix: Reproducibility Commands

To reproduce all results in a clean environment:

```
# 1. Activate environment
.venv\Scripts\activate

# 2. Run Reproducibility & Mathematical Invariant Tests
pytest -v tests/test_reproducibility.py

# 3. Re-run Grid Study (uses cached weights, runs in ~3 s)
python scripts/run_grid_study.py

# 4. Re-run Boundary Condition Extension (uses cached weights, runs in ~4 s)
python scripts/run_boundary_extension.py

# 5. Re-run Controlled Penalty Weight Sweep (uses cached weights, runs in ~8 s)
python scripts/run_penalty_sweep.py
```

All raw data are preserved in `results/grid_study.csv`, `results/boundary_extension.csv`, and `results/penalty_sweep/heat_penalty_sweep.csv`.